

- Management and deployment with familiar desktop tools and skills
- Predictable per user pricing

Amazon WorkSpaces

Amazon WorkSpaces is a managed, secure cloud desktop service. You can use it to provision either Windows or Linux desktops in just a few minutes and quickly scale to provide thousands of desktops to workers across the globe. WorkSpaces eliminates the need to procure and deploy hardware or install complex software. You can quickly add or remove users as your needs change. Users can access their virtual desktops from multiple devices or Web browsers.

Amazon WorkSpaces uses PCIO or WorkSpaces Streaming Protocol (WSP) for pixel streaming workspaces. Figure 2 shows its architecture. For authentication, we can create a standalone managed directory, or connect to a corporate on-premises directory so that existing credentials can be used to obtain seamless access to corporate resources. We can use multi-factor authentication (MFA) for additional security. We can also use AWS Key Management Service (AWS KMS) to encrypt data at rest, for disk I/O, and for volume snapshots.

Google Workspace

Google Workspace (earlier known as Google Apps and later G Suite) is a collection of cloud computing, productivity and collaboration tools, software and products developed and marketed by Google. It consists of Gmail, Contacts, Calendar, Meet and Chat for communication; Currents for employee engagement; Drive for storage; and the Google Docs suite for content creation. An Admin Panel is provided for managing users and services.

Google Workspace adds enterprise features such as custom email addresses at a domain (e.g., @yourcompany.com), an option for unlimited Drive storage, additional administrative tools and advanced settings, as well as 24/7

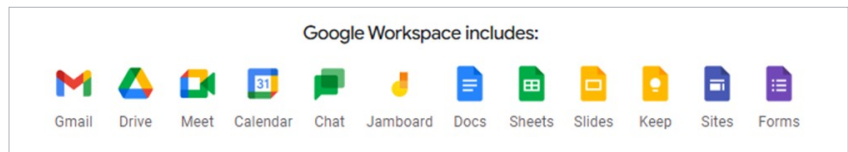


Figure 3: Google Workspace

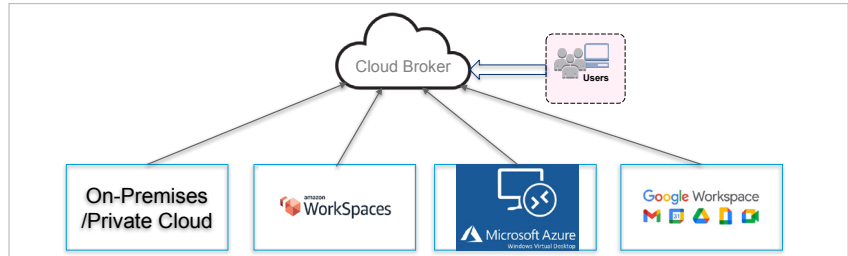


Figure 4: Hybrid/multi-cloud workspaces with a common management service

phone and email support.

It gives flexible tools that enable agility, interoperability, customisation and seamless work at scale—across companies, software, devices and distances.

Google Workspace is built with the security, reliability, and cloud infrastructure necessary to keep information, identities, applications, and devices safe. It provides encryption at-rest and in-transit, and automatic protection against sophisticated phishing attacks and emerging malware threats. It helps admins streamline authentication, asset protection, and operational control—all from one dashboard.

Multi-cloud workspaces

Enterprises may prefer hybrid or multi-cloud scenarios for various reasons — to avoid vendor lock-in, to leverage unique services, to optimise costs, etc. To enable such capability with a single management console and seamless operations, a

desktop management fabric service can be leveraged. Vendors such as Citrix, VMware and Workspot are offering such cloud broker/management fabric services.

Even though public cloud vendors relieve enterprises from infrastructure management, the latter are still responsible for their tenant management, image management, OS and applications management, user profile and data management, migration from on-premises, choice of VM instances, etc. There are many open source tools available to help enterprises perform these operations. Many of these tools are cloud vendor neutral and provide standard interfaces.

For orchestration and automation, open source tools such as Terraform, Ansible, Jenkins, Chef, etc, can be leveraged. For monitoring, open source tools such as Prometheus, Logilizer, Istio, etc, can be used. Visualisation tools such as Grafana are useful for dashboards. **END** 🐼

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